

BENJAMIN LAM

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Work Experience

Tesla, Inc.

Jan 2025 – Present

Test Engineer

- **Managing testing operations for a major drive unit program (Jira)**, responsible for test planning, design, execution, and analysis to support change validation and failure analysis
- Abuse & Inverter Fault Response Testing: Integrated **automated oscilloscope triggering** into our dynamometer test setup, achieving **~10x faster test execution** and improved data organization. **Created the first documentation for this category of testing**, defining standards and processes to guide development of future tests.
- **Led root cause analysis of a critical component failure impacting potential vehicle damage**. Created new electrical and thermal metrics to evaluate hardware and firmware solutions. Recommended firmware fix and supported validation
- Developed **Python-based data pipelines to streamline test result analysis** and easily compare engineering changes; projects include thermal modeling to quantify component power dissipation and characterizing part buckling to compare manufacturing process changes

Ceder Group (UC Berkeley)

Aug 2022 – Dec 2024

Graduate Student Researcher

- Built **automated in-situ XRD sample loader** (Flask/Python/I²C) for integration into our lab's autonomous synthesis lab [[Github](#)]
- Built custom Li-air battery test system controlled with **Flask/Python/I²C** to cycle cells in **PID-controlled** relative humidity/gas composition. Designed Swagelok-type cell for airtight sealing, assembly/control of electronic sensors and actuators, and all software. Adds **Li-air battery test capabilities to our lab** (Python, C++, JavaScript, HTML) [[Github](#)]
- Research on **degradation mechanism** of Li-NASICON solid-state Lithium conductors in alkaline environments using time-resolved **XRD, SEM/EDS, ICP-OES** experiments, data analysis performed with **MATLAB**. Results contextualized with DFT surface simulations and ab-initio Pourbaix diagrams to **uncover root cause failure mode of cell. Eliminates viability of phosphate chemistries**. [[Advanced Energy Materials \(2024\), First Author](#)]
- **SEM/EDS image processing** (MATLAB) tool enabled large dataset analysis feasible. Utilized Hough transform to identify image features

ASML

Jun 2021 – Sep 2021

Intern, Droplet Generator

- **Data analysis of in-field** tin droplet generator performance (**Python**): Developed functions to flag heat cycling anomalies, which were integrated into the new dashboard for engineers to view events indicating oxide formation and subsequent failure
- Failure analysis for droplet generator; corresponded with external lab to investigate mechanism of tin-oxide formation

Education

University of California, Berkeley

Dec 2024

MS in Materials Science Engineering, **GPA: 3.97**

- Courses: **Model predictive control**, Signals and systems, Additive Manufacturing, **Graduate Thermodynamics**, X-ray Diffraction, Materials characterization (XRD, SEM, TEM, etc.), Bonding Theory, Inorganic chemistry
- **MPC Final Project: Automated path generation and correction** for parking a 2-wheeled rover. Simulations with wheel slip considered to evaluate control robustness to external disturbances

University of California, Irvine

Jun 2022

BS in Mechanical Engineering and Materials Science Engineering (Double Major), **GPA: 3.98**

- Dean's Honor List (All quarters), Graduated Summa Cum Laude
- **Bowman Lab Research**: Custom MATLAB library developed for processing of EIS datasets, contributed to **2 papers**: [[ACS Applied Nano Materials \(2023\)](#)] [[Advanced Functional Materials \(2024\)](#)]
- **Hui Lab Research**: Developed a simulated model for a **microfluidic valve**. **Made new valve design** with truncated valve floor helps improve size variation of generated droplets (down to $\sigma = 1\%$, **3x improvement**) [[Micromachines \(2022\)](#)]
- Courses: **Materials failure, Control Systems**, Fuel cells, Solar energy

Skills

Mechanical: SolidWorks (CSWA Certified), Autodesk Fusion 360, Autodesk CFD, Basic shop experience

Electrical/Programming: Python (4+ years), MATLAB (4+ years), PCB Prototyping, Oscilloscope, High Voltage Experience, **Inverters, Motor controls**

Materials Science: XRD, SEM/EDS, ICP-OES/MS, EIS, Aqueous Chemistry, TGA/DSC, 3+ years Chemical Lab Experience